

Bluestock Mutual Fund Analytics Capstone

Data Engineering, Analytics and Dashboard Project

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Problem Statement & Objectives

Problem Statement

The mutual fund industry generates massive volumes of complex data daily, including:

- Net Asset Value (NAV) history
- Assets Under Management (AUM)
- Systematic Investment Plan (SIP) records
- Investor transaction logs

Analyzing this fragmented data manually is highly inefficient and prone to error, hindering actionable insights.

Objectives

Develop an automated, end-to-end analytics solution:

- Build a robust ETL pipeline
- Clean and validate unstructured data
- Create a centralized SQLite database
- Perform Exploratory Data Analysis (EDA)
- Calculate key performance metrics
- Build an interactive Power BI dashboard
- Generate actionable business insights

Data Sources & Scope

Data Inputs

- Fund Master Data & Benchmark Indices
- Historical NAV Data
- Assets Under Management (AUM) Data
- Monthly SIP Inflows
- Category-wise Inflows
- Investor Transaction Logs
- Portfolio Holdings

Project Scope:

Total Funds Analysed: 40

Period Covered: 2022–2025



Project Architecture Pipeline



EDA Highlights: Growth Trends

Key Industry Findings

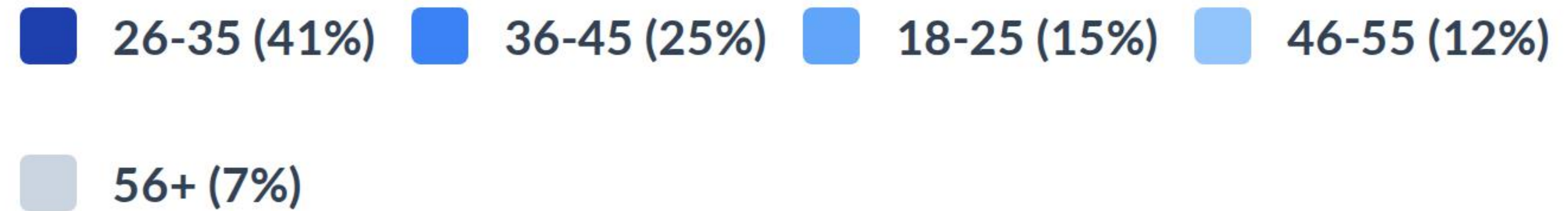
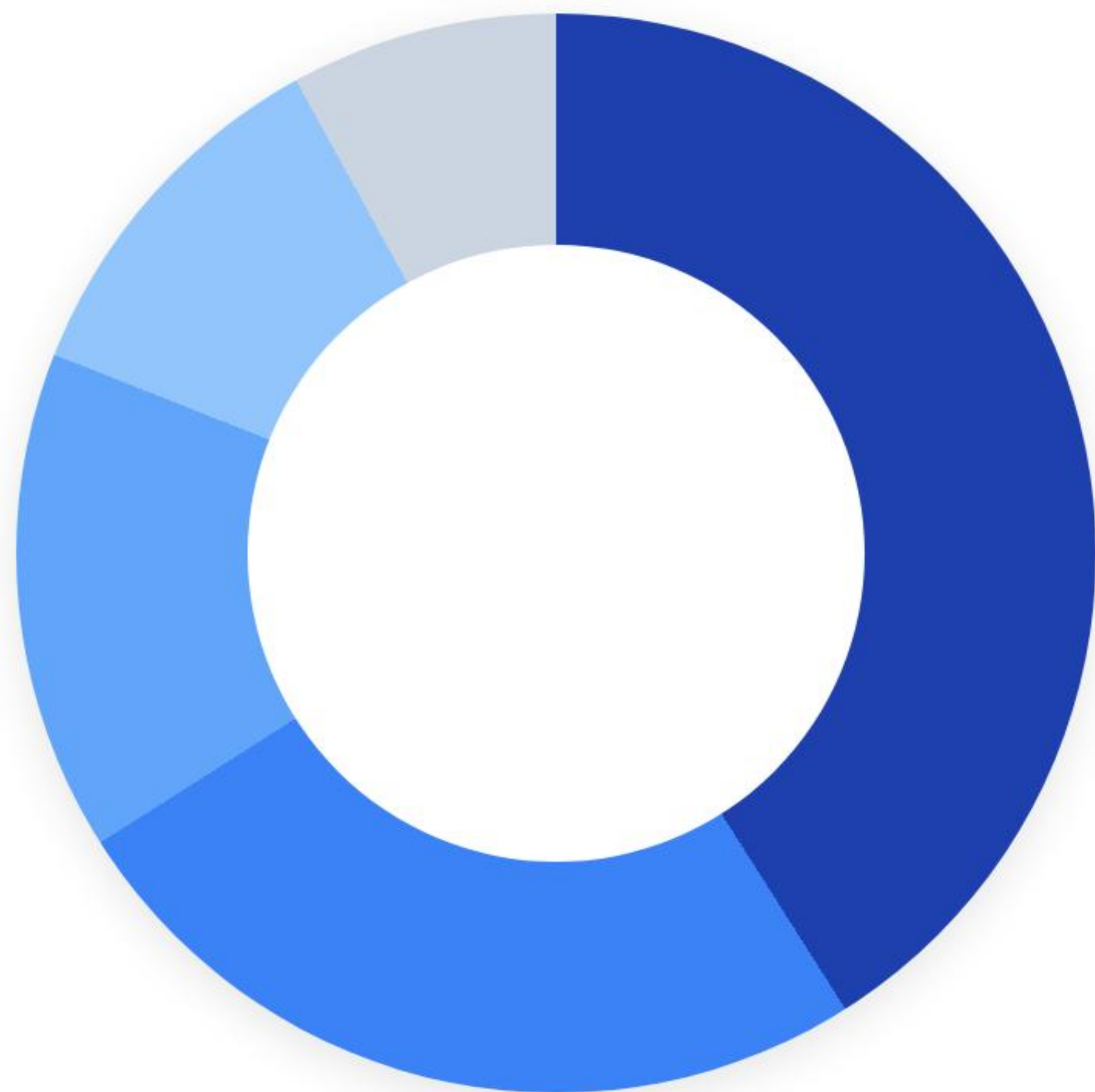
- **Industry AUM** showed significant and strong growth over the analyzed period.
- **SIP participation** increased consistently, indicating changing retail investor behavior.
- **Folio count** doubled during the study period (2022-2025).
- **Equity funds** attracted the major portion of total inflows.

Industry AUM Trend (Conceptual)



EDA Highlights: Investor Analytics

Age Group Distribution



Investor Demographics & Behavior

- Majority of investors belong to the **middle age groups** (26-45 years).
- **SIP investments** dominate the overall transaction volume.
- Strong participation observed from major economically active states.
- **T30 (Top 30) cities** contribute the most significant portion of investments.

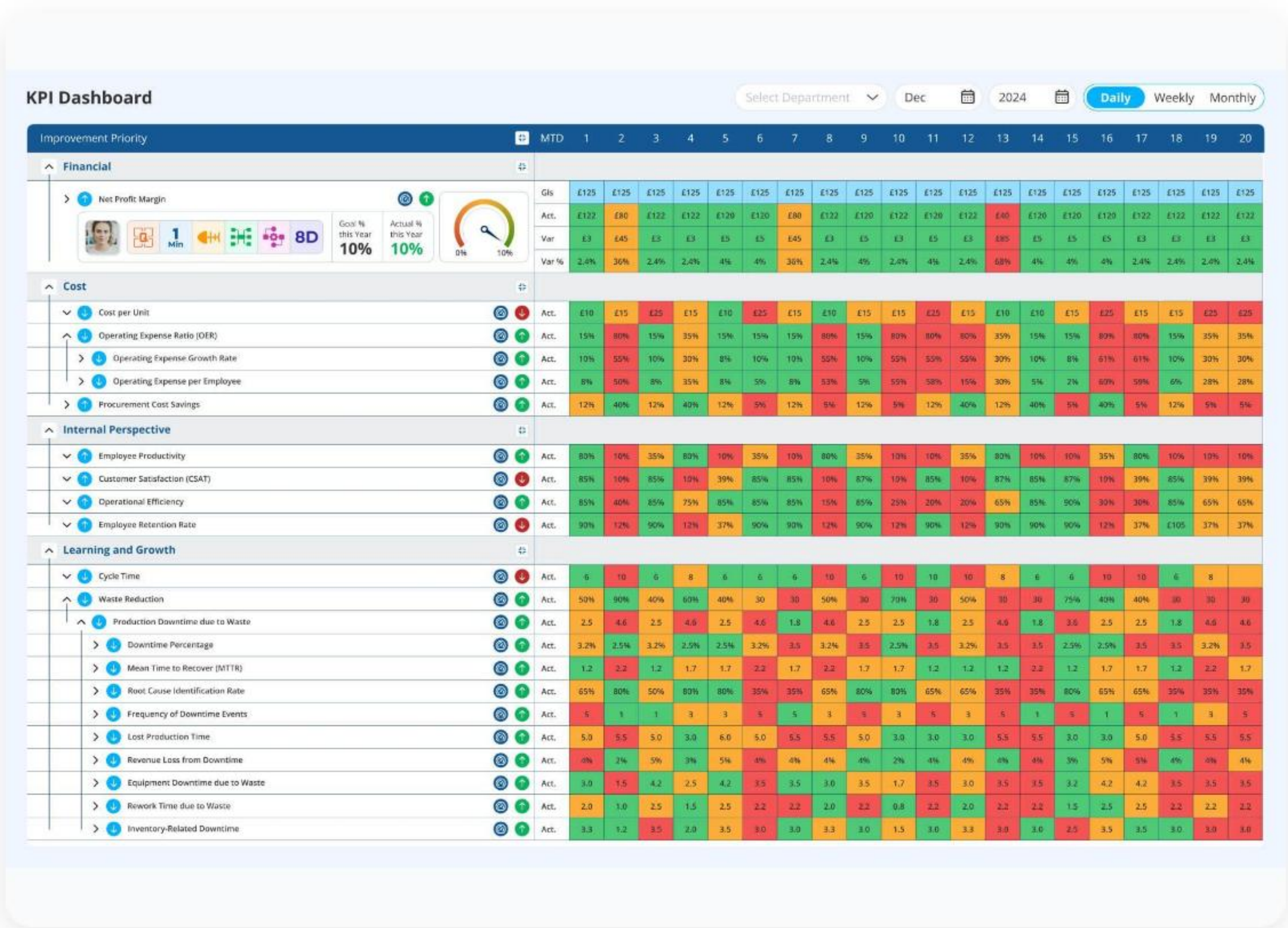
Performance Analytics Framework

Metrics Calculated

The project employs rigorous mathematical models to evaluate funds beyond absolute returns.

- **CAGR:** Compound Annual Growth Rate over 1, 3, and 5 years.
- **Sharpe Ratio:** Risk-adjusted return relative to volatility.
- **Sortino Ratio:** Focuses specifically on downside risk.
- **Alpha & Beta:** Measures performance against the benchmark.
- **Maximum Drawdown:** Historical risk indicator.

Purpose: To accurately measure and compare risk-adjusted performance across the 40 analyzed funds.

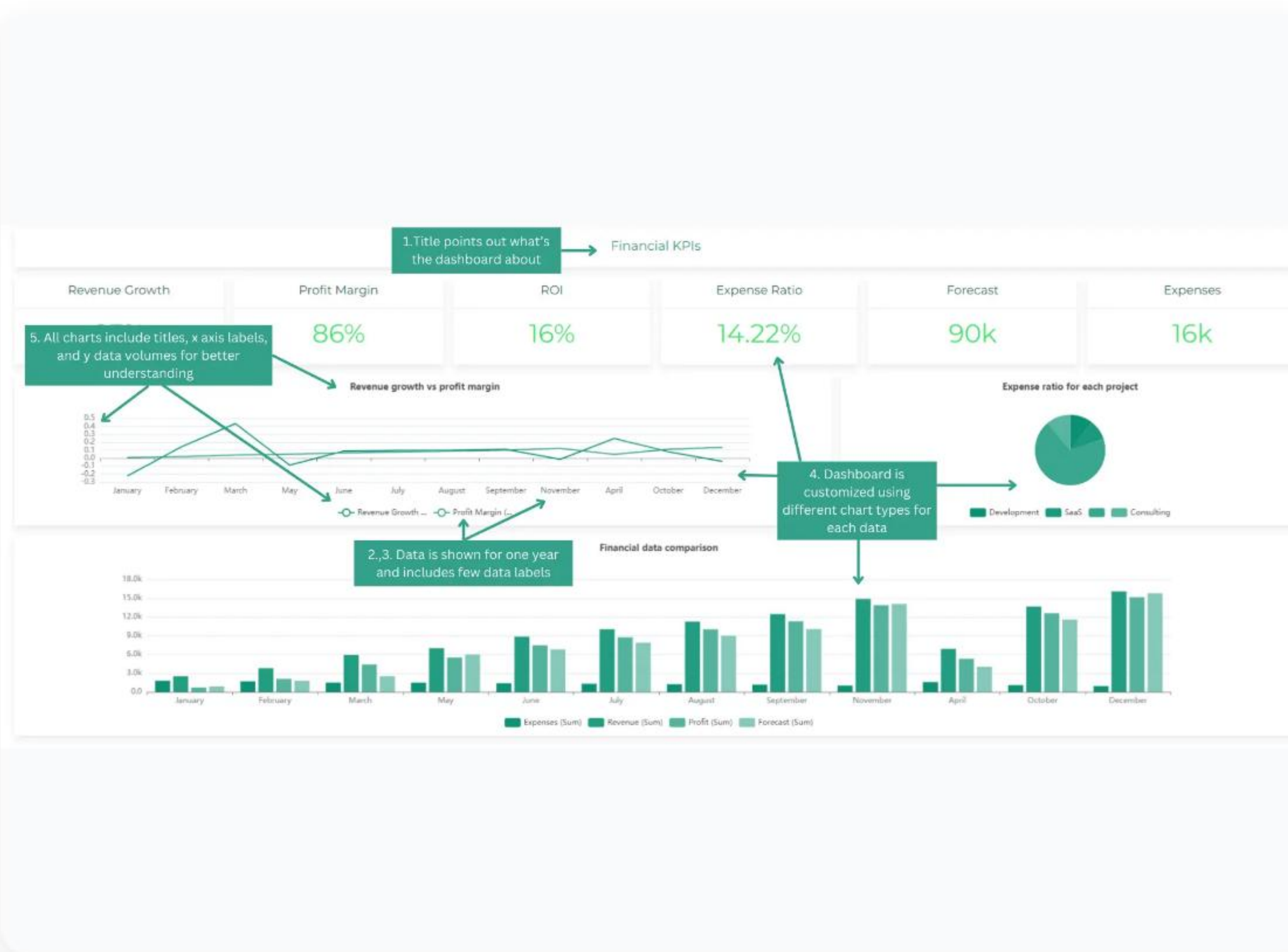


Advanced Risk Analytics

Deeper Portfolio Insights

Moving beyond basic metrics to understand complex risk factors:

- **VaR (95%):** Value at Risk calculation to estimate maximum expected loss.
- **CVaR:** Conditional Value at Risk for tail risk assessment.
- **Rolling Sharpe Ratio:** Analyzing performance consistency over time rather than static periods.
- **SIP Continuity Analysis:** Tracking investor discipline.
- **Sector HHI Concentration:** Herfindahl-Hirschman Index to measure portfolio diversification.
- **Fund Recommendation Engine:** Logic built on analyzed metrics.

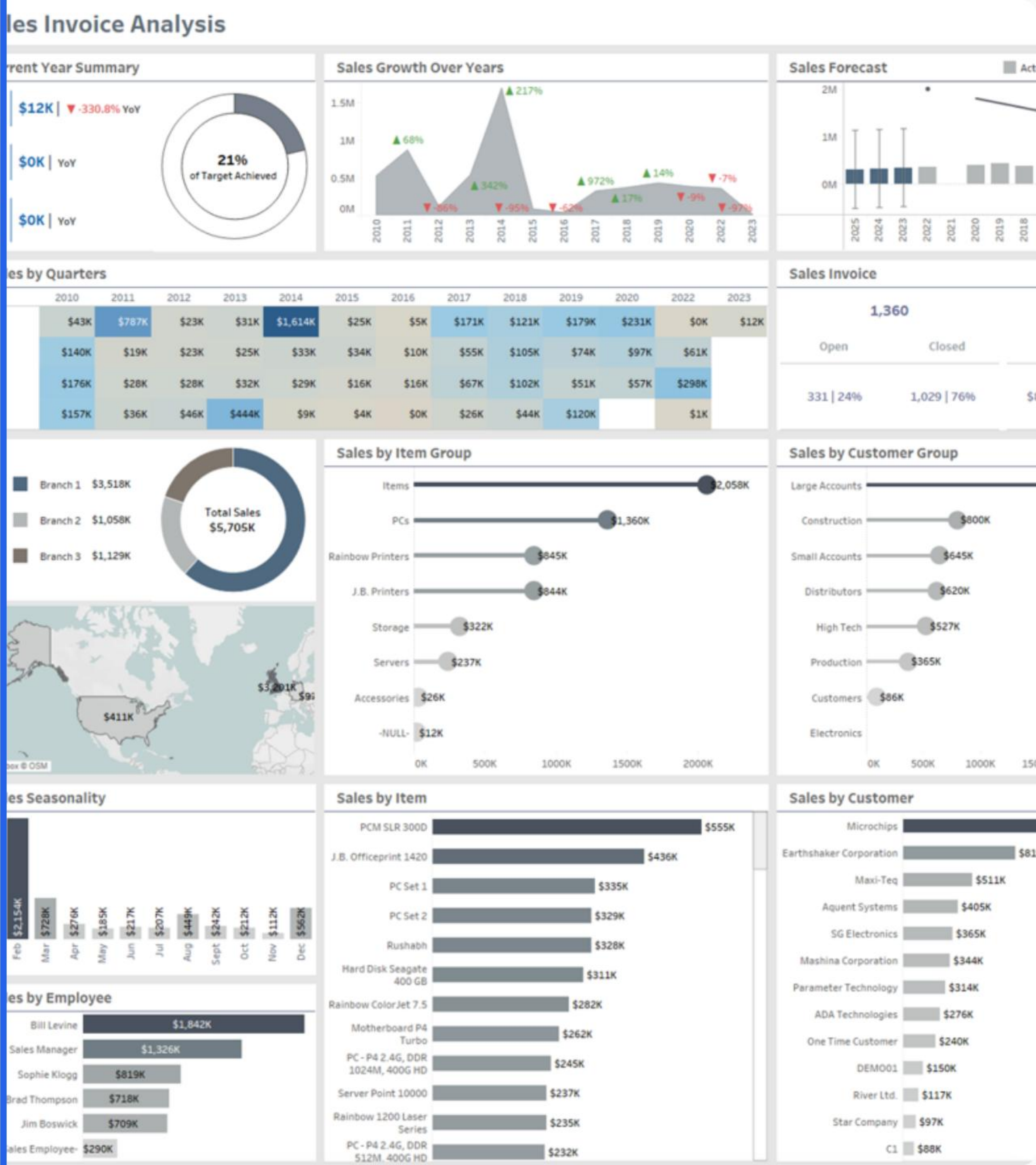


Dashboard: Overview & Performance

Interactive Analytics Interface

The primary dashboard pages provide a macro view of the industry alongside granular fund performance data.

- **Industry Overview:** High-level metrics tracking total AUM, active folios, and overall market trajectory.
- **Fund Performance:** Detailed screening of all 40 funds.
- **Interactive KPI Cards:** Dynamic top-line metrics that respond to user filters.
- **AUM Trend Analysis:** Visual representation of asset growth over the 2022-2025 period.
- **Fund Comparison Tools:** Side-by-side risk vs. return scatter plots and NAV trajectory charts.

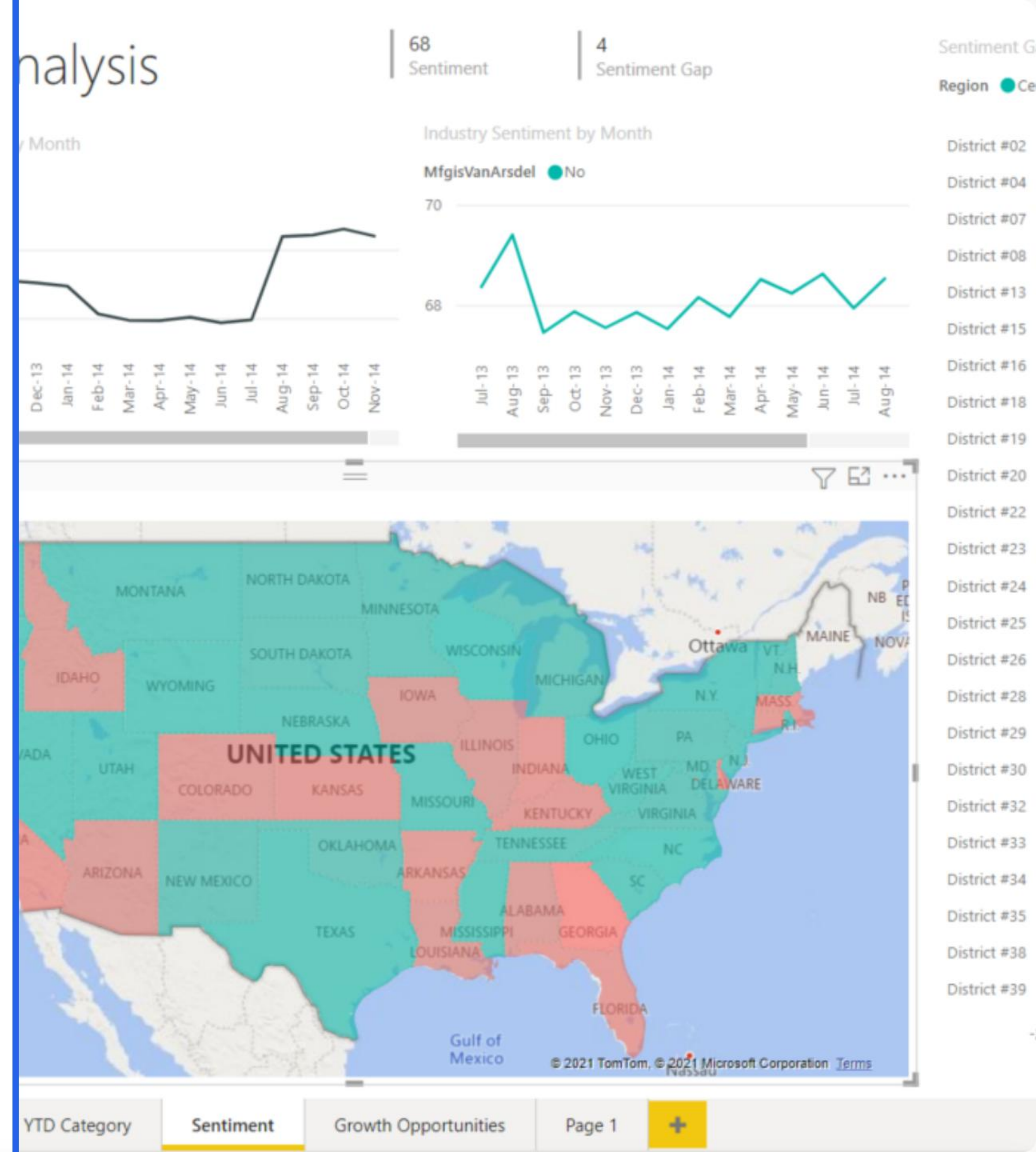


Dashboard: Investors & Markets

Demographic & Flow Analysis

The secondary dashboard pages delve into investor behavior and category-specific money flows.

- **Investor Analytics:** Breakdown of the investor base by age, gender, and investment size.
- **SIP & Market Trends:** Monitoring the stability and growth of systematic investments.
- **Geographic Analysis:** Visualizing transaction amounts originating from different states and city tiers.
- **Transaction Behaviour:** Comparing SIP, Lumpsum, and Redemption volumes.
- **Category Flow Monitoring:** Heatmaps tracking where capital is moving (e.g. Equity vs. Debt)



Conclusion

Key Findings

- **Industry AUM** continues to exhibit strong, sustained growth.
- **SIP participation** remains highly resilient regardless of market volatility.
- Risk-adjusted performance (Sharpe/Sortino) differs significantly among top-tier funds.
- Investor behaviour varies notably by geographic region and age demographic.

Recommendations

- Focus marketing efforts on promoting **consistent SIP investing** among younger demographics.
- Recommend **diversified funds** to mitigate identified sector concentration risks.
- Continuously monitor **risk-adjusted returns** rather than absolute NAV growth.
- Track portfolio concentration (HHI) to avoid overlap in investor portfolios.

Thank You

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 Questions & Discussion

Image Sources



https://static.vecteezy.com/system/resources/previews/020/459/722/non_2x/abstract-cloud-computing-technology-concept-internet-data-storage-database-server-cyber-technology-background-illustration-vector.jpg

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